

APARAJITO SAHA

rajsaha@umich.edu — LinkedIn — Website

RESEARCH INTERESTS

3D perception & scene understanding, instruction-following mobile manipulation, reinforcement learning for skill coordination

EDUCATION

Ph.D. Robotics, University of Michigan, Ann Arbor
Advisor: Dr. Bernadette Bucher

Ann Arbor, MI, USA
Aug 2026 — Exp. May 2030

M.S. Robotics, University of Michigan, Ann Arbor
Cumulative GPA: 4.00/4.00

Ann Arbor, MI, USA
Aug 2024 — May 2026

B.S. Electrical & Computer Engineering + Computer Science, Cornell University
Cumulative GPA (double major): 3.74/4.00

Ithaca, NY, USA
Aug 2018 — May 2022

RESEARCH EXPERIENCE

Graduate Student Research Assistant, Mapping & Motion Lab
Principal Investigator: Dr. Bernadette Bucher

University of Michigan
Dec 2024 — Present

- Formulated the *interactive object search* problem, in which a mobile manipulator must reason about necessary physical interactions with an unexplored indoor environment to eliminate occlusions, identify visual affordances and locate semantically-defined target objects
- Designed a robotic task benchmark for the interactive object search problem to evaluate geometric and semantic reasoning for high-level planning alongside fine-grained control capabilities for mobile manipulator systems
- Led implementation efforts for a photorealistic, physics-driven extension on Nvidia's IsaacSim platform that leverages procedural generation and 3D asset population for creating multi-room indoor scenes for interactive object search (**W1**)
- Advised a team of graduate and undergraduate students for deploying state-of-the-art perception and manipulation systems such as MAST3R-SLAM, RoboPoint and AnyGrasp on a Boston Dynamics Spot robot, with the objective of producing an open-source software stack for instruction-following mobile manipulation
- Developed a method to add object-map alignment to visual-semantic frontier mapping, providing robust localization and open-set object goal navigation capabilities

Undergraduate Research Assistant, EmPRISE Lab
Principal Investigator: Dr. Tapomayukh Bhattacharjee

Cornell University
Aug 2021 — May 2022

- Engineered a prototype circuit board for managing real-time communication of tactile feedback from a whole-arm fabric skin sensor system, earning an acknowledgment for the contribution in B. Xu et. al. "CushSense: Soft, Stretchable, and Comfortable Tactile Sensing Skin for Physical Human-Robot Interaction" *International Conference on Robotics & Automation*, 2024

PROFESSIONAL EXPERIENCE

Software Development Engineer, Amazon Fulfillment Technologies & Robotics
Manager: Todd Clemetson

North Reading, MA, USA
Jul 2022 — Aug 2024

- Designed autonomous robotic movement solutions deployed to over 1 million mobile robotic drive units in Amazon's global warehouse network
- Led a zero-downtime serverless rearchitecture and migration effort for a critical production service responsible for computing and updating unavailable resources on warehouse floors
- Contributed to algorithms for storage and retrieval optimization used in Amazon's new Sequoia fulfillment centers, reducing delivery miss rates
- Developed a continuous improvement and deployment pipeline framework for the robotic movement team, leading to a 2 hour reduction in downtime and maintenance per service per week across 8 Tier-1 and Tier-2 services
- Participated in on-call rotations to resolve high-severity events, mentored interns and new engineers on engineering practices and projects, and held open office hours for operations managers to troubleshoot floor issues

PUBLICATIONS

Workshop Papers

W1 Aparajito Saha, Zhen Hao Gan, Jinjia Guo, Jacob Skwirs, Jeremy Acheampong, Anton Arapin, Chahyon Ku, Yue Hu, Nima Fazeli, Bernadette Bucher. *PhyGS: Physically-Grounded Controllable Scene Generation*. Workshop on Multi-Agent Embodied Intelligent Systems, **CVPR** 2026.

Technical Reports

TR1 Aparajito Saha*, Amulya Khurana*, Krithik Ranjan*. *AroundSound: Digital Echolocation for the Visually Impaired*. CircuitCellar Issue 386, Sept. 2022. (* denotes equal contribution)

TEACHING

Graduate Student Instructor, University of Michigan, Ann Arbor
EECS 442: Computer Vision (240 students/ semester)

Ann Arbor, MI, USA
Aug 2025 — Dec 2025

Teaching Assistant, Cornell University
ECE 5725: Design with Embedded Operating Systems (80 students/ semester)

Ithaca, NY, USA
Aug 2021 — May 2022

Peer Advisor, Cornell University
ENGRG 1050: Engineering Seminar (25 students/ semester)

Ithaca, NY, USA
Aug 2021 — May 2022

HONORS

Robotics Department Fellowship, University of Michigan, Ann Arbor
Bachelor of Science, Cum Laude, Cornell University
Departmental Honors in Computer Science, Cornell University
Dean's List (Engineering), Cornell University

2026 — 2027
2022
2022
2018 — 2022

MENTORSHIP

University of Michigan, Ann Arbor

- Jinjia Guo, M.S. Robotics *Sept 2025 — Dec 2025*
- Kornvik Tanpipat, M.S. Robotics *Sept 2025 — Dec 2025*
- Saumit Vedula, M.S. Robotics *Sept 2025 — Dec 2025*
- Aileen Yan, M.S. Electrical & Computer Engineering *Sept 2025 — Dec 2025*
- Chenrui Gao, B.S. Computer Science & Engineering *Sept 2025 — Dec 2025*
- Georgia Zender, B.S. Computer Science & Engineering *Feb 2025 — May 2025*

Amazon Fulfillment Technologies & Robotics

- Hemant Palavali, Software Development Engineer *December 2023 — March 2024*

Cornell University

- Cornell Cup Robotics Team (20+ students) *Aug 2021 — May 2022*

SKILLS

Programming Languages: Python, C++, Java, MATLAB

Software: IsaacSim, PyTorch, ROS2, Slurm, Git, Docker, AWS Lambda, AWS DynamoDB

Hardware: NVIDIA Jetson, Raspberry Pi, Rapid Prototyping (3D Printing, Soldering, Circuit Board Design)